

"تدریس برق ماشین"

"آرکسین بزرگوار"

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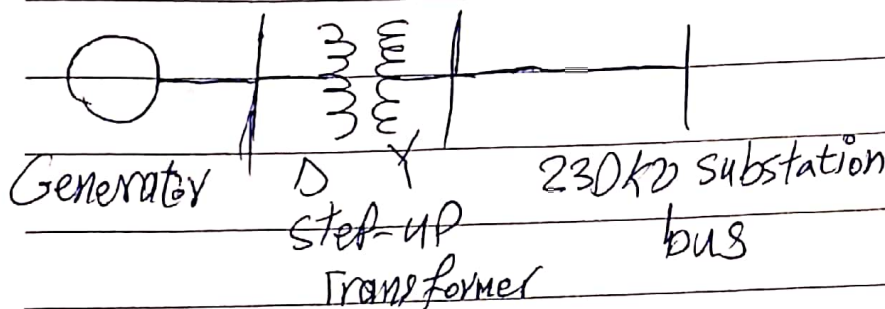
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Generator: 250 MVA, 15.75 kV, $X_G = 16\% = 0.16 \text{ pu}$

Transformer: 300 MVA, 15.75 kV / 230 kV, $X_T = 10\% \text{ pu}$

Line: 200 km, $X_L = 0.35 \frac{\Omega}{\text{km}}$



$$S_b = 250 \text{ MVA}, V_{b,1} = 15.75 \text{ kV} \Rightarrow V_{b,2} = V_{b,1} \times \frac{230}{15.75} = 230 \text{ kV}$$

$$\Rightarrow X_{T, \text{pu}} = X_T \times \frac{250}{300} = 10\% \times \frac{250}{300} = \frac{1}{3} \times \frac{10}{100} = \frac{1}{3} \times \frac{1}{10} = \frac{1}{30} = 0.0333 \text{ pu}$$

$$\Rightarrow X_{L, \text{actual}} = 0.35 \frac{\Omega}{\text{km}} \times 200 \text{ km} = 70 \Omega$$

$$\Rightarrow Z_{\text{base}} = \frac{V_{b,2}^2}{S_b} = \frac{(230 \text{ kV})^2}{250 \text{ MVA}} = \frac{52900}{250} = 211.6 \Omega$$

$$X_{L, \text{pu}} = \frac{X_{L, \text{actual}}}{Z_{\text{base}}} = \frac{70}{211.6} \approx 0.331 \Rightarrow X_{th, \text{pu}} = X_{G, \text{pu}} + X_{T, \text{pu}} + X_{L, \text{pu}}$$

$$\Rightarrow X_{th, \text{pu}} = 0.16 + 0.0333 + 0.331 = 0.5243 \text{ pu}$$

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$$S = 100 \text{ MVA}$$

base

الف) ۳

$$V_{base,1} = 10 \text{ kV}$$

و تارهای انتقال Bus 1 به Bus 2 و 3 و 4 و 5

$$V_{base,2} = V_{base,1} \times \frac{100 \text{ kVA}}{10 \text{ kV}} = 100 \text{ kV}$$

و تارهای انتقال در Bus 3 و 4 و 5

$$V_{base,3} = V_{base,2} \times \frac{100 \text{ kVA}}{10 \text{ kV}} = 100 \text{ kV}$$

و تارهای انتقال در Bus 4 و 5

$$S_{load} = 100 \text{ MVA}$$

$$X_{series,pu} = X_{T1} + X_{line} + X_T = 0.1 + 0.12 + 0.12 = 0.34 \text{ pu} \Rightarrow S_{load,pu} =$$

$$\frac{100 \text{ MVA}}{100 \text{ MVA}} = 1.0 \text{ pu} \Rightarrow P.F. = 0.9 \Rightarrow P = 1.0 \times 0.9 = 0.9 \text{ pu}$$

$$\Rightarrow Q = 1.0 \times 0.433 = 0.433 \text{ pu}$$

Bus 3, Bus 4 و 5

$$V_1 = 1 \angle 0^\circ \Rightarrow I = \frac{V_1 - V_2}{jX_{series}}, S = V_2 I^* = P + jQ$$

$$b = -P/X_{series} = -0.9/0.34 = -2.647 \Rightarrow a - (a+b) = Q/X_{series}$$

$$= 0.433/0.34 = 1.273 \Rightarrow a + b = 1.273 \Rightarrow a - (-2.647) = 1.273 \Rightarrow a = -1.374$$

$$b = 0.433/0.34$$

$$a - (-1.374) = 0 \Rightarrow a = -1.374$$

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$$\Rightarrow |V_E| = \sqrt{a^2 + b^2} = \sqrt{0.185^2 + 0.111^2} = \sqrt{0.0342 + 0.0123} = \sqrt{0.0465} = 0.216 \text{ pu}$$

$$\Rightarrow \angle V_E: \delta = \tan^{-1}\left(\frac{b}{a}\right) = \tan^{-1}\left(\frac{-0.111}{0.185}\right) = -11.21^\circ \Rightarrow V_E =$$

$$V_E = 0.216 \angle -11.21^\circ \text{ pu}$$

$$\text{Bus 2: } I = \frac{V_1 - V_E}{jX_{\text{series}}} = \frac{1 - 0.216 \angle (-11.21^\circ - j \sin 11.21^\circ)}{j0.25}$$

$$\Rightarrow V_E = 0.216 \angle (-11.21^\circ - j \sin 11.21^\circ) = 0.185 \angle -11.21^\circ - j0.111$$

$$\Rightarrow V_1 - V_E = (1 - 0.185 \angle -11.21^\circ) + j0.111 = 0.814 + j0.111 \Rightarrow I =$$

$$\frac{0.814 + j0.111}{j0.25} = \frac{0.814 + j0.111}{j0.25} = 3.256 - j0.444 \text{ pu}$$

$$\text{Bus 2: } V_2 = V_1 - jIX_T = 1 - j(3.256 - j0.444) \times 0.12$$

$$\Rightarrow jI = j(3.256 - j0.444) = 0.444 + j3.256 \Rightarrow jIX_T = 0.12(0.444 + j3.256)$$

$$= 0.0533 + j0.3907 \Rightarrow V_2 = (1 - 0.0533) - j0.3907 = 0.9467 - j0.3907$$

$$\Rightarrow |V_2| = \sqrt{0.9467^2 + 0.3907^2} = \sqrt{0.8962 + 0.1526} = \sqrt{1.0488} = 1.024 \text{ pu}$$

$$\Rightarrow \theta = \tan^{-1}\left(\frac{-0.3907}{0.9467}\right) = -21.04^\circ$$

✓

$$-j\omega(4 \text{ S})x_{0,0.15} \Rightarrow jI_{x_{0,0.15}} = (0.4 \text{ S} + j\omega(4/10))x_{0,0.15} = 0.4x_{0,0.15}$$

$$+j90010 \Rightarrow V_r = (0.92294 - j0.01012) - j(0.0114 + 0.10010) = 0.91895 \angle$$

$$\Rightarrow |V_P| = \sqrt{0.15^2 + 0.001^2} = \sqrt{0.15^2 + 0.001^2} = \sqrt{0.15^2} = 0.15 \text{ V}$$

$$\Rightarrow \theta_r = \tan^{-1} \left(\frac{-0.0901}{0.9154} \right) = -5.41^\circ \Rightarrow \text{Bus 2: } 0.9154 \angle -5.41^\circ$$

BUS 3: 0A19^{Pu}~~2~~-2191°

Bus 401, Nov 25 - 11, 21

Bus 5: 01/10/2017 ^{Pa} - 11/2017

ب) وندار سن ۱۸۵۸ تا ۱۹۰۱ با اسیر و نیست است اسیران ها گری

$$X_{T_1} = 0,115, X_{line} = 0,0150, X_{T_2} = 0,115$$

$X_{total} = 0, \text{ kata } P_u$

یک بانک خارجی به جای اتصال دسکد، Bus 1 و $\sim K^v$ قرار دادیم. زیرا قرار

کتاب ۱۴۰۰، ص ۱۵۰، مباحث و مسائل در باره سیستم های انتقال توان در Bus ۳ و
 $|V_2| = |V_E| \Rightarrow V_3 = 1.05 \text{ pu}, \theta_3 = -3.74^\circ$

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حال راکتو از Bus 3 به Bus 1: $(X_{T1} + X_{line} = 9.5 + 0.12 = 0.1295 \text{ pu})$

$$Q_{r1} = \frac{V_{r1}^2 Q_r - V_{r1}^2}{X_{T1} + X_{line}} = \frac{1.1 \text{ kV}^2 \times 0.99 \text{ pu} - 1.1 \text{ kV}^2}{0.1295} = -2.907 \text{ pu}$$

حال راکتو از Bus 3 به Bus 1 (از نظر سرنه در حالت بار)

$$Q_{rE} = 0.121522 \text{ pu} \Rightarrow Q_{r1} + Q_{rE} + V_{r1}^2 B_c = 0 \Rightarrow B_c = -\frac{Q_{r1} + Q_{rE}}{V_{r1}^2}$$

$$= \frac{-(-2.907) + 0.121522}{1.1 \text{ kV}^2} = \frac{2.818478}{1.21 \text{ pu}} = 2.33 \text{ pu}$$

$$Z_{base} = \frac{(\sum 100 \times 10^3)^2}{20 \times 10^9} = 2500 \Omega \Rightarrow B_{actual} = \frac{B_c}{Z_{base}} = \frac{2.33}{2500}$$

$$f = 60 \text{ Hz} \Rightarrow \omega = 2\pi f = 376.99 \text{ rad/s} \Rightarrow C = \frac{B_{actual}}{\omega} = \frac{0.94 \times 10^{-6}}{376.99}$$

$$\approx 2.49 \times 10^{-9} \text{ F} = 2.49 \text{ nF}$$

$G_1, G_2: 10 \text{ MVA}, 0.6 \text{ kVA}, X = 1\%$

$G_3: 10 \text{ MVA}, 3.81 \text{ kV}, X = 50\%$

$T_1: 50 \text{ MVA}, 0.3 \text{ kV} / 0.6 \text{ kV}, X = 10\%$

$T_2: 10 \text{ MVA}, 364 \text{ kV} / 2.2 \text{ kV}, X = 15\%$

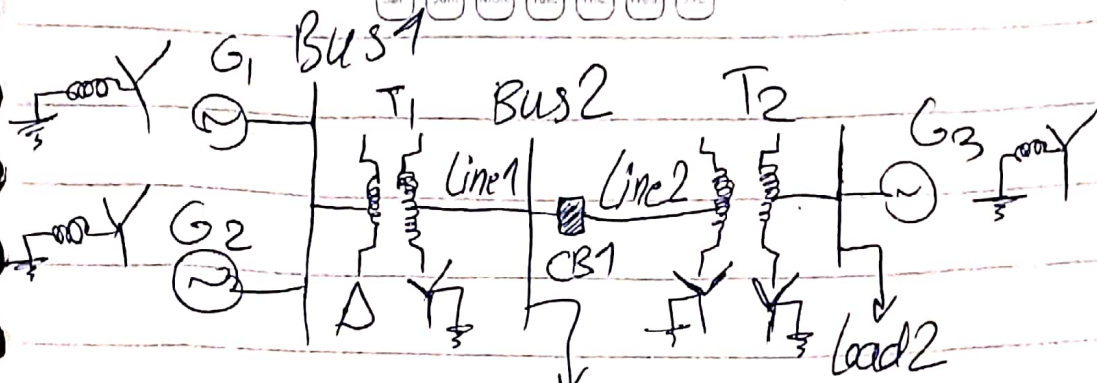
Line 1: 20 MVA, 0.6 kV, PF = 0.9 lagging

" 2: 6 MW, 3.81 kV, PF = 0.9 lagging

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$$\begin{array}{l|l|l}
 S_b = 100 \text{ MVA} & S_b = 100 \text{ MVA} & S_b = 100 \text{ MVA} \\
 V_b = 9.4 \text{ kV} & V_b = 9.4 \text{ kV} \times \frac{44}{9.1} = 44.18 \text{ kV} & V_b = 9.4 \text{ kV} \times \frac{44}{9.1} \times \frac{1}{\sqrt{3}} = 25.14 \text{ kV} \\
 & Z_b = \frac{V_b}{S_b} = 129.1 \Omega & = 5.14 \text{ kV}
 \end{array}$$

$$X_{g1} = X_{g2} = 0.1 \times \frac{S_{b(\text{new})}}{S_{b(\text{old})}} = 0.1 \times \frac{100}{10} = 0.1$$

$$X_{gr} = 0.1 \times \left(\frac{V_{b(\text{old})}}{V_{b(\text{new})}} \right)^2 \times \frac{S_{b(\text{new})}}{S_{b(\text{old})}} = 0.1 \times \left(\frac{9.1}{25.14} \right)^2 \times \frac{100}{10} = 0.14$$

$$X_{t1} = 0.1 \times \left(\frac{V_{b(\text{old})}}{V_{b(\text{new})}} \right)^2 = 0.1 \times \left(\frac{9.1}{25.14} \right)^2 = 0.14$$

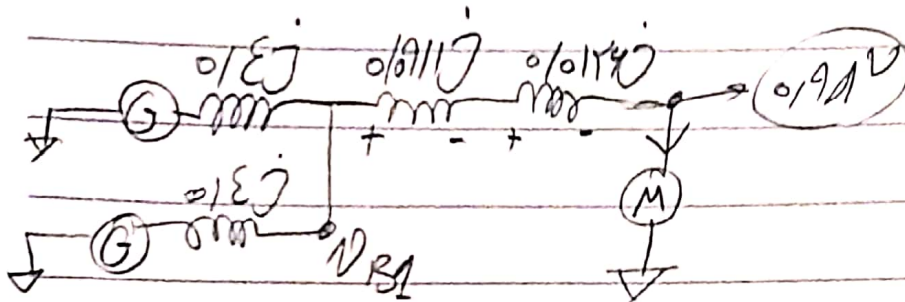
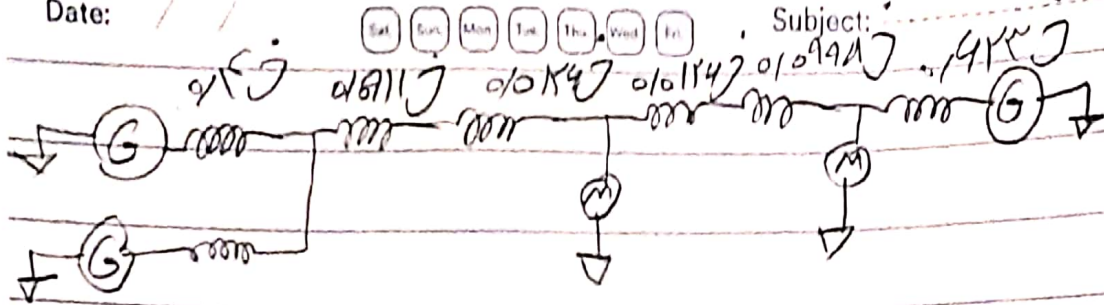
$$X_{tr} = 0.1 \times \left(\frac{V_{b(\text{old})}}{V_{b(\text{new})}} \right)^2 \times \frac{S_{b(\text{new})}}{S_{b(\text{old})}} = 0.1 \times \left(\frac{9.1}{25.14} \right)^2 \times \frac{100}{10} = 0.14$$

$$X_{\text{line1}} = X_{\text{line2}} = \frac{1}{129.1} = 0.0077$$

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$$S = VI^* \Rightarrow I^* = \frac{S}{V} \Rightarrow S = \frac{V}{Z} + \frac{1}{Z}j \Rightarrow I = \frac{V}{Z} + \frac{1}{Z}j$$

$$= 0.41 - 0.2j / V_{B1} = 0.91 + (0.41 - 0.2j)(0.15j)$$

$$= 1.0229 - 0.0102j \Rightarrow E_{g1} = E_{g2} = V_{B1} + 0.15j(0.41 - 0.2j)$$

$$= 1.25V + 0.162j$$

V
(10)